Noise-Attenuated Legged Locomotion via Noise-Proxy-Informed Residual Actions

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***Abstract*—Legged robots that operate in human-centric indoor environments must not only track velocity commands robustly, but also minimize the acoustic disturbance caused by their repetitive foot–ground impacts. Standard locomotion reinforcement learning prioritizes tracking and stability, while relatively little attention has been given to the acoustic consequences of foot–ground interactions. We propose *NALOCO*, a noise-attenuating residual adaptation framework that uses proprioceptive observation history to infer a mechanically motivated noise proxy, reducing impact-induced acoustic noise without retraining the nominal locomotion controller. *NALOCO* consists of: 1) a residual policy-based controller, 2) a five-term noise proxy function defined in simulation, and 3) a novel noise proxy estimator network (NPNet) to estimate the proxy function from proprioceptive observation history. This architecture enables real-world deployment without privileged sensors. The residual policy is conditioned on the NPNet latent representation and produces bounded action corrections while using Parseval regularization to moderate its sensitivity to input variations. It is optimized with a velocity-adaptive quiet-walking reward that applies stronger pressure for quieter touchdowns at low commanded speed and gradually relaxes that pressure as speed rises. We validate *NALOCO* on Unitree Go2, B2, and G1 in simulation and on five real-world terrains, where activating the residual policy reduces peak SPL by up to $9.6\,\mathrm{dB}$ relative to the reference policy and by up to $16.2\,\mathrm{dB}$ relative to the built-in MPC, while preserving velocity tracking performance.**

***Index Terms*—Reinforcement Learning, Legged Locomotion, Noise Attenuation, Residual Policy, Noise Proxy.**

I. Introduction

REINFORCEMENT learning (RL) has become a powerful framework for learning legged locomotion policies that track velocity commands, traverse rough terrain, generate diverse gait patterns, and recover from external disturbances. These RL-based controllers have demonstrated robust performance, particularly in scenarios where physical environments or system dynamics are difficult to model accurately [1–4]. In recent years, legged robots have evolved beyond standalone locomotion, with RL enabling the joint learning of diverse capabilities, such as agile traversal, manipulation, and navigation, that were previously developed as separate modules [5–7]. As a result, RL-based approaches are increasingly being considered as viable alternatives or complements to these modular pipelines, reflecting a broader shift toward data-driven control paradigms.

Despite these advances, existing RL approaches have primarily focused on *robot-centric* objectives such as command tracking and robust locomotion, while complementary considerations such as acoustic disturbance and accelerated mechanical wear are largely underexplored. In particular, the repetitive foot–ground impacts in robust gaits generate significant acoustic noise, which can limit practical robot deployment in *human-centric* environments. However, addressing this acoustic noise without compromising locomotion performance remains challenging because it entails modifying foot–ground contact behavior in ways that tend to degrade velocity tracking accuracy [8–11].

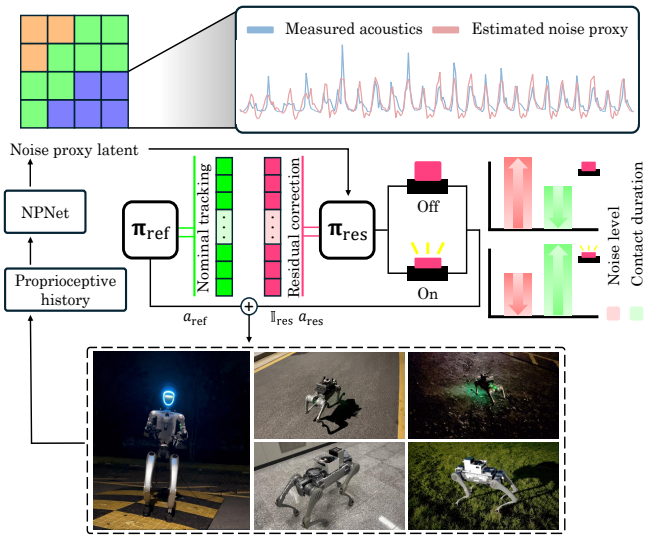


Fig. 1. A pretrained reference locomotion policy provides nominal joint actions, while the residual policy—conditioned on a proprioceptive noise proxy latent from NPNet—applies residual corrections that attenuate impact-driven contact noise without compromising velocity tracking performance.

Conventional simulation-based RL environments for legged locomotion do not directly provide acoustic observations of foot–ground interactions. Prior studies have generally addressed contact-induced acoustic noise indirectly by incorporating contact-related quantities, such as contact force and foot velocity, or gait patterns into the reward objective. Although these methods can promote less impact-intensive contact behavior, they tend to provide the controller with limited awareness of the acoustic consequences of the current locomotion state. Moreover, noise reduction is embedded directly into the optimization of the locomotion policy. Even when a competent locomotion policy already exists, this requires whole-policy retraining or fine-tuning at the risk of catastrophic forgetting of previously learned locomotion skills. This coupling limits policy reuse and can compromise command tracking performance, making it difficult to reduce contact-induced noise while preserving the performance of a pretrained locomotion controller.

In this letter, we propose *NALOCO*, **N**oise-**A**ttenuated

legged **LOCO**motion via noise-proxy-informed residual actions, an RL framework that attenuates contact-induced acoustic noise while preserving the competence of a pretrained locomotion controller (Fig. 1). Instead of retraining an entire policy under a coupled quietness objective, *NALOCO* freezes a reference locomotion policy for nominal actions and learns only a residual correction dedicated to noise attenuation. Rather than learning this residual policy from proprioception alone, we enrich the policy with an auxiliary latent that encodes contact-noise-relevant structure of the current locomotion state. This latent is obtained by training a noise proxy estimator network (NPNet) to estimate our designed noise proxy function from proprioceptive observation history, with body velocity estimation as an auxiliary task.

Our main contributions are as follows:

- **Residual noise attenuation:** We decouple noise attenuation from nominal locomotion by learning a residual policy over a pretrained locomotion controller, enabling quiet contact behavior without whole-policy retraining while preserving velocity-tracking competence.
- **Noise-proxy-informed NPNet:** We define a five-term contact noise proxy function for simulation-side supervision and train a noise proxy estimator network, NPNet, to estimate it from proprioceptive observation history, providing a deployable auxiliary latent that promotes quiet locomotion through residual actions in real-world deployment.
- **Morphology-agnostic training pipeline:** We validate the proposed framework on both quadrupedal and humanoid robots within a unified system, demonstrating cross-morphology applicability and scalability across heterogeneous robotic embodiments.

II. Related Work

A. Contact-Induced Acoustic Noise

In legged locomotion, acoustic noise is generated primarily at foot–ground contact events. Each contact converts the kinetic energy of the descending foot into a short impact, producing large contact forces and high-frequency mechanical transients [12]. These transients radiate as airborne sound and, in indoor settings, also propagate as structure-borne vibration through floors and surrounding structures. The emitted acoustic level depends on the severity and temporal characteristics of the impact, including the magnitude and duration of the contact force [13], pre-contact foot velocity, particularly its vertical component [14, 15], joint acceleration at impact [16], and the stepping rate imposed by the gait [8]—as well as on the compliance of the foot and terrain. Consequently, gaits that favor firm, high-frequency contacts for robustness tend to be acoustically intrusive even when velocity tracking remains accurate. Most learning-based locomotion controllers are trained under *robot-centric* objectives such as command tracking, disturbance rejection, and fall prevention [1–3, 17–19]. Within these objectives, hard contacts are often tolerated or even reinforced because they improve traction and recovery, making impact-driven noise an unintended byproduct of competent locomotion rather than an explicit failure mode.

B. Noise-Aware Legged Locomotion

In *human-centric* environments such as hospitals, libraries, offices, and residential spaces, low acoustic disturbance is crucial [20, 21]. To mitigate this acoustic noise, previous work on quiet legged locomotion primarily reduces emissions by shaping gait and contact behavior. Yu et al. [8] develop a quiet and stable guide-dog controller with lower stepping frequency and smoother swing motion for blind and low-vision users, motivated by the need to avoid acoustic and motion disturbance during close-proximity assistance. Cao et al. [10] learn a quiet gait for indoor quadrupeds by penalizing pre-contact foot velocity according to an estimated gait phase, reducing acoustic noise under a quietness–agility trade-off. These studies establish that stepping rate, foot trajectories, and contact timing strongly influence perceived noise, but they are often tied to specific gait designs, applications, or platforms.

Other learning-based approaches incorporate noise-related conditioning and impact estimation into locomotion policy learning. QuietPaw [9] introduces a Conditional Noise-Constrained Policy that conditions behavior on a noise-reduction level and uses critic-side value representation decomposition so that one policy can span a continuum of agility–noise preferences without retraining. QuietWalk [11] targets humanoid quiet walking under diverse footwear. An inverse-dynamics-constrained physics-informed network estimates vertical ground reaction forces from proprioception and supplies an impact penalty during RL training, removing the need for force sensors at deployment. These approaches demonstrate that noise-aware objectives improve acoustic comfort, yet they typically couple quietness with the locomotion policy itself.

Gait-shaping approaches [8, 10, 16] redesign contact behavior for a particular setting, limiting reuse across tasks and morphologies. QuietPaw keeps a single end-to-end policy and trades agility against noise through preference conditioning, rather than preserving a high-performance locomotion prior. QuietWalk introduces a physically meaningful impact cue for humanoid footwear variation. However, it still optimizes quiet walking within the base locomotion policy and relies solely on vertical ground reaction force, which captures only part of the contact dynamics relevant to acoustic noise generation.

In contrast, *NALOCO* freezes a pretrained reference controller for nominal tracking and learns only residual actions for noise attenuation [22, 23]. The residual is conditioned on NPNet's proprioceptive multi-term noise proxy latent and shaped by a velocity-adaptive quiet reward. This formulation targets quieter contact without whole-policy retraining, while remaining applicable across quadruped and humanoid embodiments under one pipeline.

III. NALOCO

Our goal is to learn a legged locomotion policy that preserves robust velocity tracking while explicitly attenuating impact-driven footstep noise. *NALOCO* mainly achieves this by (i) composing control as a bounded residual over a frozen pretrained locomotion policy, (ii) defining a five-term contact noise proxy function in simulation, (iii) training NPNet to

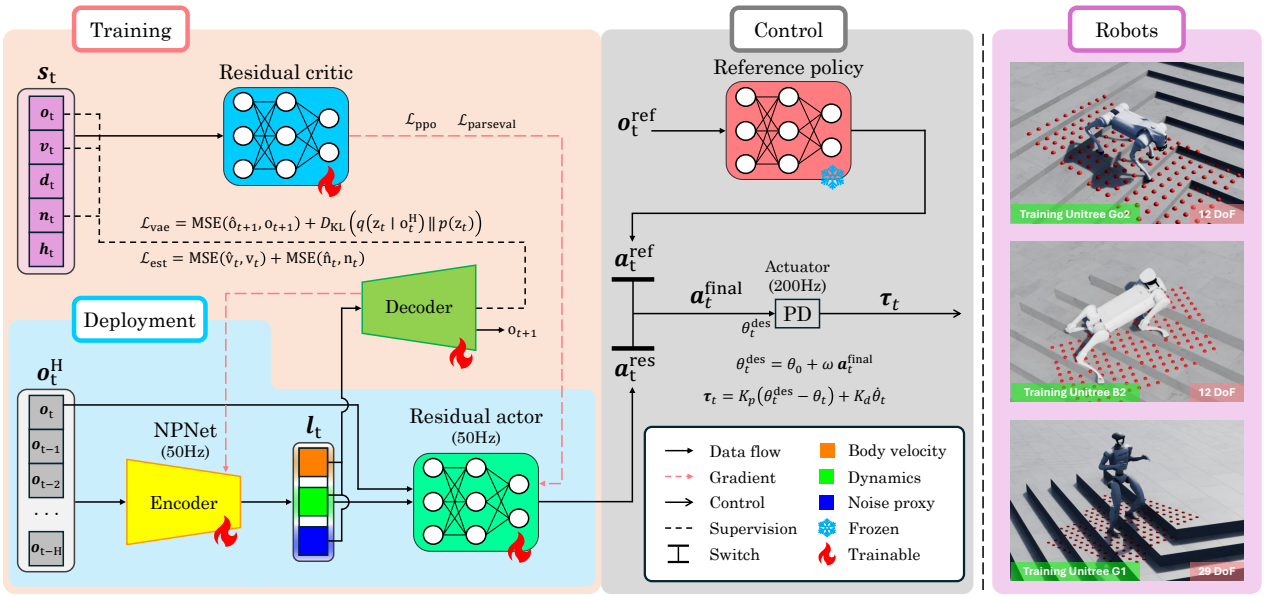


Fig. 2. **Overview of NALOCO.** A frozen reference policy produces nominal actions a_t^{ref} and the residual actor outputs a correction a_t^{res}. The composed action is $a_t^{\text{final}} = a_t^{\text{ref}} + \mathbb{I}_{\text{res}}\, a_t^{\text{res}}$, where $\mathbb{I}_{\text{res}}$ is a binary switch that activates or deactivates the residual action, recovering the reference policy when disabled.

infer the proxy from proprioceptive observation history, and (iv) optimizing the residual policy with a velocity-adaptive quiet reward and Parseval-regularized PPO (Fig. 2).

A. Problem Formulation

We model legged locomotion as a partially observable Markov decision process (POMDP). At each control step the agent receives a proprioceptive observation o_t and outputs joint-position offsets a_t relative to a default robot pose. The objective is to maximize the expected discounted return

$$J(\pi) = \mathbb{E}\big[\sum_t \gamma\, r_t\big], \tag{1}$$

where $\gamma \in (0,1)$ is the discount factor, and r_t is the reward at timestamp t. We use an asymmetric actor–critic [24]: the actor receives only deployable proprioception—body angular velocity, projected gravity, velocity commands, joint states, and previous action—while the critic additionally observes privileged information—body linear velocity, disturbance forces, noise proxy and a height scan—during training. A proprioception history $o_t^{\text{H}} = (o_t, o_{t-1}, \ldots, o_{t-\text{H}})$ is maintained for NPNet. The same observation and action structure applies to the quadrupeds and the humanoid, with morphology-dependent dimensions.

B. Residual Control

Let π_{ref} denote a frozen pretrained reference policy that already provides competent locomotion, and o_t^{ref} denote the observation of the reference policy. The residual formulation is independent of how π_{ref} was trained. The nominal action is $a_t^{\text{ref}} = \pi_{\text{ref}}(o_t^{\text{ref}})$ and, rather than conditioning the residual on a_t^{ref}, we condition it on the current observation o_t, and the NPNet's latent vector ℓ_t (Section III-D):

$$a_t^{\text{res}} = \alpha\, \tanh\big(f_\theta([o_t, \ell_t])\big), \qquad a_t^{\text{final}} = a_t^{\text{ref}} + \mathbb{I}_{\text{res}}\, a_t^{\text{res}}, \tag{2}$$

where α, f_θ are the residual scale and the residual actor, respectively. Four mechanisms keep the residual as a bounded correction rather than a second competing controller: (i) tanh bound, (ii) near-zero initialization of the residual output layer, (iii) Parseval orthonormality regularization on residual hidden weights (Section III-F), and (iv) latent decoupling, so that ℓ_t is detached from the policy gradient and NPNet is updated only by its auxiliary estimation and prediction loss.

C. Noise Proxy Function

Direct acoustic observation is unavailable in conventional simulators. We therefore define a privileged five-term noise proxy function from contact and kinematic quantities computable in simulation, analogous to injecting a thermal actuator model so that temperature—unobserved in standard physics engines—can be used as a policy observation and reward [25]. Let $\mathcal{F}$ be the set of feet, K be the temporal offset, and $\mathbb{I}_{t,i}^{\text{fc}} \in \{0,1\}$ indicate first contact of foot i at timestep t. With contact force $\mathbf{F}_{t,i}$, foot velocity $\mathbf{v}_{t,i}$ and their planar and vertical components $\mathbf{v}_{t,i}^{xy}$ and $\mathbf{v}_{t,i}^{z}$, and joint accelerations $\ddot{\mathbf{q}}_t$, we form

$$n_t^{\text{impact}} = \sum_{i\in\mathcal{F}} \|\mathbf{F}_{t,i}\|_2\, \mathbb{I}_{t,i}^{\text{fc}}, \tag{3}$$

$$n_t^{\text{impulse}} = \sum_{i\in\mathcal{F}} \Big(\sum_{t}^{t+K} \|\mathbf{F}_{t,i}\|_2\Big) \mathbb{I}_{t,i}^{\text{fc}}, \tag{4}$$

$$n_t^{\text{acc}} = \|\ddot{\mathbf{q}}_t\|_2^2, \tag{5}$$

$$n_t^{\text{xy}} = \sum_{i\in\mathcal{F}} \|\mathbf{v}_{t,i}^{xy}\|_2^2\, \mathbb{I}_{t,i}^{\text{fc}}, \tag{6}$$

$$n_t^{z-} = \sum_{i\in\mathcal{F}} \text{ReLU}\big(-\mathbf{v}_{t-1,i}^{z}\big)^2\, \mathbb{I}_{t,i}^{\text{fc}}, \tag{7}$$

and compress them as

$$\mathbf{n}_t = \log\big(1 + [n_t^{\text{impact}}, n_t^{\text{impulse}}, n_t^{\text{acc}}, n_t^{\text{xy}}, n_t^{z-}]\big) \in \mathbb{R}^5. \tag{8}$$

The impact term in Eq. (3) captures the instantaneous contact-force magnitude at first contact, whereas the impulse term in Eq. (4) captures the accumulated contact-force magnitude over a temporal window at the contact step. The joint-acceleration term in Eq. (5) captures rapid kinematic transients that can accompany impacts and abrupt load redistribution. The planar-velocity term in Eq. (6) captures tangential foot motion at first contact, serving as a proxy for sliding or scuffing. For the vertical velocity term in Eq. (7), we use the foot's vertical velocity from the previous control step to capture its downward motion immediately before first contact. The $\mathrm{ReLU}(-v^z_{t-1,i})$ retains only downward motion, so upward swing does not contribute. The foot-wise terms are aggregated across all feet, while the joint-acceleration term is computed over the full joint state, yielding a fixed five-dimensional representation whose dimension remains morphology-independent.

D. NPNet: Proprioceptive Noise Proxy Estimator

NPNet is a β-VAE [26, 27] that encodes proprioception history $\mathbf{o}^{\mathrm{H}}_t$ into a latent representation $\ell_t \in \mathbb{R}^{40}$. Specifically, ℓ_t is decomposed into three stochastic latent variables:

$$\ell_t = [\hat{\mathbf{v}}_t, \mathbf{z}_t, \hat{\mathbf{n}}_t], \quad \hat{\mathbf{v}}_t \in \mathbb{R}^3,\ \mathbf{z}_t \in \mathbb{R}^{32},\ \hat{\mathbf{n}}_t \in \mathbb{R}^5. \tag{9}$$

This decomposition encourages the latent space to capture distinct aspects of locomotion: instantaneous body linear velocity $\hat{\mathbf{v}}_t$, dynamics context $\mathbf{z}_t$, and a noise proxy for locomotion-related acoustic information $\hat{\mathbf{n}}_t$. Each component is inferred through a dedicated stochastic head from the shared encoder representation. A decoder reconstructs the next observation $\hat{\mathbf{o}}_{t+1}$ from ℓ_t. Let $\mathbf{v}_t$ denote privileged body velocity, $\mathbf{o}_{t+1}$ privileged next observation, $q(\mathbf{z}_t \mid \mathbf{o}^{\mathrm{H}}_t)$ the approximate posterior, $p(\mathbf{z}_t)$ a standard normal prior, and $\mathbf{n}_t$ a noise proxy function. The training objective combines supervised regression of velocity and noise proxy, next observation reconstruction, and a KL regularizer on $\mathbf{z}_t$:

$$\begin{aligned}\mathcal{L}_{\mathrm{NPNet}} = {} & w_{\mathrm{vel}}\|\hat{\mathbf{v}}_t - \mathbf{v}_t\|_2^2 + w_{\mathrm{recon}}\|\hat{\mathbf{o}}_{t+1} - \mathbf{o}_{t+1}\|_2^2 \\ & + w_{\mathrm{KL}}\, D_{\mathrm{KL}}\big(q(\mathbf{z}_t \mid \mathbf{o}^{\mathrm{H}}_t) \,\|\, p(\mathbf{z}_t)\big) + w_{\mathrm{n}}\|\hat{\mathbf{n}}_t - \mathbf{n}_t\|_2^2,\end{aligned} \tag{10}$$

where w_{vel}, w_{recon}, w_{KL}, and w_{n} are scalar loss weights. At policy inference, NPNet uses the means of the inferred Gaussian posteriors from the stochastic heads.

We further use AdaBoot [3] to bootstrap the body velocity and noise proxy estimates. Early in training the estimates are unreliable and become more accurate as training progresses. AdaBoot mixes $\hat{\mathbf{v}}_t$ with privileged velocity and $\hat{\mathbf{n}}_t$ with a noise proxy function during rollouts, with probability

$$p_{\mathrm{boot}} = 1 - \tanh(\mathrm{CV}(\mathbf{R})), \tag{11}$$

where $\mathrm{CV}(\mathbf{R})$ is the coefficient of variation of episode returns $\mathbf{R}$. As returns stabilize, p_{boot} approaches 1, and the residual actor increasingly relies on NPNet, matching real-world deployment.

E. Reward Design

We use the locomotion reward of [3], comprising command tracking and regularization, and augment it with a velocity-adaptive quiet reward and a noise proxy penalty:

$$r_t = r^{\mathrm{track}}_t + r^{\mathrm{reg}}_t + r^{\mathrm{quiet}}_t + r^{\mathrm{noise}}_t. \tag{12}$$

The quiet reward r^{quiet}_t is driven by a scalar Q_t that, at each valid touchdown, combines a logarithm of contact-force magnitude with pre-contact downward foot speed, using the same cues as n^{impact}_t and n^{z-}_t.

$$Q_t = \sum_{i\in\mathcal{F}} \Big(w_F \log(1 + \|\mathbf{F}_{t,i}\|) + w_{v_z}\,\mathrm{ReLU}(-v^z_{t-1,i}) \Big) \mathbb{I}^{\mathrm{valid}}_{t,i}, \tag{13}$$

where w_F and w_{v_z} are positive weights and $\mathbb{I}^{\mathrm{valid}}_{t,i}$ marks a first contact of foot i with sufficient force. The quiet reward is expressed as:

$$r^{\mathrm{quiet}}_t = w_t \exp\big(-(Q_t - Q^\star)^2/\sigma^2\big), \tag{14}$$

where w_t, $Q^\star$, and σ are the velocity-dependent weight, the target, and the kernel width, respectively. To avoid overly conservative gaits at high speed, we adapt its weight with commanded planar speed v^{cmd}_t using an interpolation coefficient $\lambda_t \in [0,1]$:

$$\lambda_t = \mathrm{clip}\Big(\frac{v^{\mathrm{cmd}}_t - v_{\mathrm{low}}}{v_{\mathrm{high}} - v_{\mathrm{low}}}, 0, 1\Big), \quad w_t = (1-\lambda_t)w_{\mathrm{low}} + \lambda_t\, w_{\mathrm{high}}, \tag{15}$$

where v_{high}, v_{low}, w_{high}, and w_{low} are scalar weights. So w_t interpolates from w_{low} at v_{low} to w_{high} at v_{high}, increasing quiet pressure at low commanded speeds and relaxing it as speed rises. Also, we construct the noise proxy penalty r^{noise}_t as a sum of negative-weighted penalties on each proxy term in Eqs. (3)–(7), so that reducing each term increases the return.

F. Training Objective and Regularization

We optimize the residual policy with the proximal policy optimization (PPO) [28] surrogate objective and adaptive KL scheduling. The total loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{surr}} + c_v\mathcal{L}_V - c_e\,\mathcal{H} + \lambda_P\,\mathcal{L}_{\mathrm{Parseval}}, \tag{16}$$

where $\mathcal{L}_{\mathrm{surr}}$ is the clipped policy surrogate, $\mathcal{L}_V$ is the value-function regression loss, $\mathcal{H}$ is policy entropy, and c_v, c_e, λ_P are scalar coefficients. Parseval regularization [29] is applied to each hidden weight matrix $W \in \mathbb{R}^{m\times n}$ of the residual actor. It penalizes the deviation of the Gram matrix from the identity on the smaller side of W,

$$\mathcal{L}_{\mathrm{Parseval}} = \sum_W \begin{cases} \|WW^\top - I\|_F^2 & m \le n, \\ \|W^\top W - I\|_F^2 & m > n, \end{cases} \tag{17}$$

encouraging approximately 1-Lipschitz layers and thereby limiting the sensitivity of the residual policy to its inputs [30].

IV. Experiments

We leverage DreamWaQ [3], a proprioceptive locomotion controller with an implicit state estimator, as a reference policy and train a residual policy with NPNet on top of it. Unless otherwise specified, NPNet is trained with all five noise-proxy terms in Eq. (8).

TABLE I
PPO AND NPNET HYPERPARAMETERS.

	Parameter	Value
Network architecture	Residual actor/critic	[512, 256, 128]
	NPNet encoder/decoder	[256, 128] / [128, 256]
Learning parameters	Epochs / mini-batches	5 / 4
	Learning rate	0.001
	Clip (ϵ)	0.2
	Discount factor (γ)	0.99
	GAE (λ)	0.95
	Adaptive KL	0.01
	Value loss coefficient (c_v)	1.0
	Entropy coefficient (c_e)	0.01
	Gradient norm clip	1.0
	Initial noise std	1.0
	Parseval weight (λ_P)	0.01
System configuration	Iterations / Episode length	5000 / 20 s
	Simulation dt	0.005 s (200 Hz)
	Control dt	0.02 s (50 Hz)

TABLE II
COMPARISON OF MEAN TOUCHDOWN IMPACT FORCE, TRACKING ERROR, AND IMPACT REDUCTION ACROSS COMMANDED VELOCITIES IN MUJOCO.

Robot	v_x	Impact [N]		Track [m/s]		Red. [%]
	[m/s]	w/o res.	w/ res.	w/o res.	w/ res.	
Go2	0.2	34.2	26.1	0.04	0.03	23.68%
	0.4	53.0	41.2	0.03	0.03	22.26%
	0.6	87.1	67.9	0.03	0.03	22.04%
	0.8	96.3	75.2	0.03	0.02	21.91%
	1.0	108.6	93.2	0.04	0.03	14.18%
B2	0.2	367.1	126.0	0.08	0.03	65.68%
	0.4	547.9	197.6	0.07	0.03	63.94%
	0.6	457.4	288.4	0.06	0.04	36.95%
	0.8	572.0	404.7	0.07	0.04	29.25%
	1.0	598.8	541.0	0.06	0.04	9.65%
G1	0.2	143.9	113.1	0.10	0.08	21.40%
	0.4	138.2	119.9	0.08	0.08	13.24%
	0.6	165.7	136.8	0.06	0.05	17.44%
	0.8	186.1	118.5	0.11	0.08	36.32%
	1.0	182.0	140.3	0.16	0.12	22.91%

A. Experimental Setup

1) Hardware Settings: We used the Unitree Go2 and B2 quadrupeds and the 29-DoF G1 as the humanoid robot. Their nominal masses are $15.21\,\mathrm{kg}$, $83.50\,\mathrm{kg}$, and $35.11\,\mathrm{kg}$, respectively. All robots are mounted with an Intel NUC for policy inference and controls. The residual policy runs synchronously at $50\,\mathrm{Hz}$ with NPNet, and the actuator PD controller runs at $200\,\mathrm{Hz}$.

2) RL Environments: We utilized Isaac Lab [31] to train NPNet and residual policy with 4096 parallel environments, and randomized various physical properties of the robot and terrains based on the randomization scheme described in [3]. All policies were trained for 5k iterations on a workstation equipped with an AMD Ryzen Threadripper PRO 7965WX CPU and an NVIDIA GeForce RTX 5090 GPU, requiring approximately 2.5 hours per robot. Shared PPO and NPNet hyperparameters are listed in Table I.

B. Evaluation Metrics

For real-world experiments, we quantify acoustic noise using sound pressure level (SPL). The audio is recorded using a DJI Mic 2 at a sampling rate of $16\,\mathrm{kHz}$. We calculate root-mean-square (RMS) amplitude of the audio signals, denoted by p^{RMS}, and compute the corresponding SPL as in [9]:

$$\mathrm{SPL}\,(\mathrm{dB}) = 20\log_{10}\left(\frac{p^{\mathrm{RMS}}}{p_0}\right), \tag{18}$$

where $p_0{=}20\,\mu\mathrm{Pa}$ is the reference sound pressure [32]. For each audio clip, we report the peak and mean SPL, using the peak as the primary metric since footstep noise is impulsive [10]. Microphone placement and recording settings were kept consistent across comparisons, and the reported acoustic levels are intended for relative comparison.

C. Simulations

1) Impact Force and Tracking Error: We evaluate the touchdown impact force and command-velocity tracking error through sim-to-sim transfer in MuJoCo [33]. The commanded forward speeds are $v_x \in \{0.2, 0.4, 0.6, 0.8, 1.0\}\,\mathrm{m/s}$ on flat terrain. For each robot and command-velocity, we conduct 20 independent trials for both the residual-off and residual-on settings. At each control step, we detect the first contact of each foot and measure the corresponding touchdown impact force as the sum of the contact-force magnitudes. The velocity-tracking error is computed using the mean-absolute-error (MAE). The results are averaged and summarized in Table II. **Go2.** Impact reduction is 23.68% at $0.2\,\mathrm{m/s}$, from 34.2 to $26.1\,\mathrm{N}$, and remains near 22% through $0.8\,\mathrm{m/s}$, from 96.3 to $75.2\,\mathrm{N}$. **B2.** The largest reductions occur at low speed, reaching 65.68% from 367.1 to $126.0\,\mathrm{N}$ at $0.2\,\mathrm{m/s}$ and 63.94% from 547.9 to $197.6\,\mathrm{N}$ at $0.4\,\mathrm{m/s}$. **G1.** Impact reduction reaches 36.32% at $0.8\,\mathrm{m/s}$, from 186.1 to $118.5\,\mathrm{N}$, and 22.91% at $1.0\,\mathrm{m/s}$, from 182.0 to $140.3\,\mathrm{N}$. These results demonstrate that across all three morphologies, the residual reduces impact forces at every commanded speed relative to the reference policy, while matching or improving velocity tracking performance.

2) Gait Contact Patterns: Under the same trials, we measure contact counts against v_x and foot-wise gait contact patterns over $5\,\mathrm{s}$ for both the residual-off and residual-on settings in MuJoCo. Fig. 3(a) presents the average contact counts of each robot, and Fig. 3(b) and Fig. 3(c) show the foot-wise contact patterns at $0.5\,\mathrm{m/s}$. **Go2.** Contact counts drop significantly from 96 to 53 at $0.4\,\mathrm{m/s}$, from 81 to 38 at $0.6\,\mathrm{m/s}$. The average contact durations are $151\,\mathrm{ms}$ for residual-off and $191\,\mathrm{ms}$ for residual-on. **B2.** The descent of foot counts is observed: from 55 to 33 at $0.6\,\mathrm{m/s}$ and from 48 to 34 at $1.0\,\mathrm{m/s}$. The average contact durations are $238\,\mathrm{ms}$ for residual-off and $338\,\mathrm{ms}$ for residual-on. **G1.** The count falls from 38 to 27 at $0.4\,\mathrm{m/s}$ and from 52 to 24 at $0.6\,\mathrm{m/s}$. The average contact durations are $238\,\mathrm{ms}$ for residual-off and $294\,\mathrm{ms}$ for residual-on. Consequently, the residual-on setting exhibits fewer ground contacts and longer contact durations across the velocities, without introducing additional steps.

3) Latent Space Preservation: To examine the consistency of the latent representations under the reference and residual policies, we visualize their latent distributions using t-SNE. Fig. 4 shows the t-SNE embeddings of the latent representations under forward, backward, lateral, and rotational com-

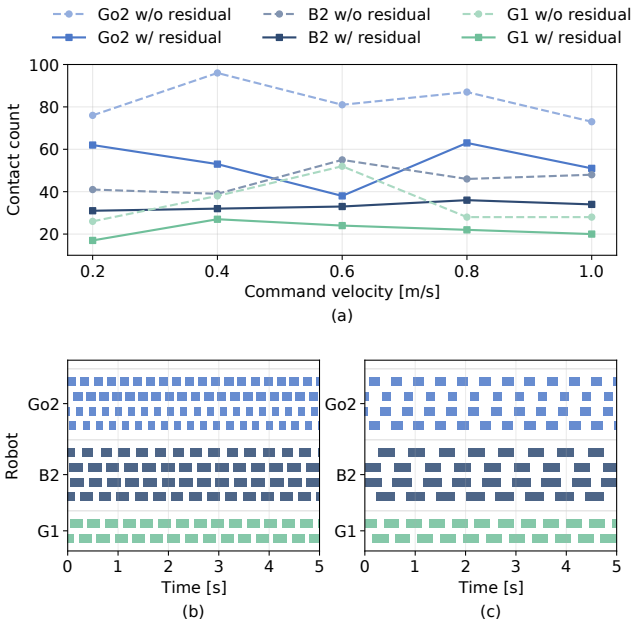


Fig. 3. (a) Comparison of contact counts across robots. (b) and (c) Foot-wise contact patterns at 0.5m/s with the residual off and on, respectively.

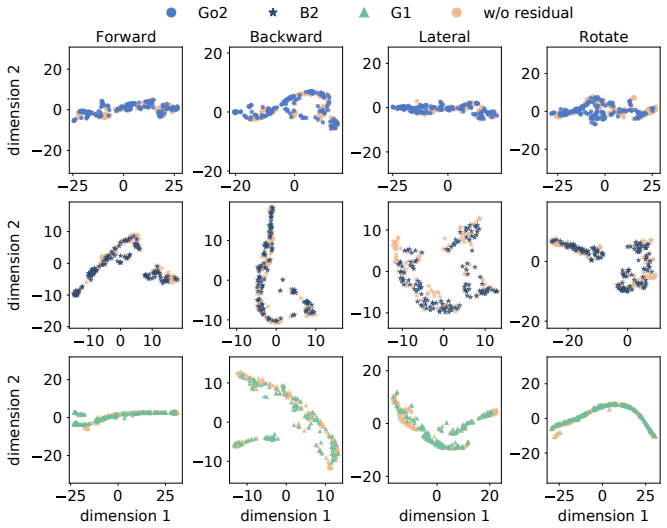


Fig. 4. Distribution of the decoded NPNet noise-proxy representation at a fixed nominal command input. Rows correspond to Go2, B2, and G1, and columns to forward, backward, lateral, and rotational commands. Residual transients and isolated outliers are excluded to focus the comparison on steady locomotion.

mands. The nominal command magnitudes are fixed at $1.0\,\mathrm{m/s}$ for linear motions and $1.0\,\mathrm{rad/s}$ for rotational motions. In addition to the visualization, we quantify residual-induced changes in the latent spaces. For each robot–motion pair, each channel is standardized using the pooled mean and standard deviation computed from both residual-off and residual-on samples. Let $(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and $(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$ denote the resulting mean and covariance for residual off and on, respectively. We report the standardized mean shift $D_\mu = \|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|_2$, total-variance ratio $R_{\mathrm{var}} = \mathrm{tr}(\boldsymbol{\Sigma}_1)/\,\mathrm{tr}(\boldsymbol{\Sigma}_0)$, and Gaussian 2-Wasserstein distance

$$D_{\mathrm{W2}}^2 = D_\mu^2 + \mathrm{tr}\left(\boldsymbol{\Sigma}_0 + \boldsymbol{\Sigma}_1 - 2\left(\boldsymbol{\Sigma}_0^{1/2}\boldsymbol{\Sigma}_1\boldsymbol{\Sigma}_0^{1/2}\right)^{1/2}\right). \quad (19)$$

Across the twelve robot–motion pairs, the mean values are D_μ=0.397, D_{W2}=0.766, and R_{var}=0.644. The robot-wise mean D_{W2} values are 0.577, 0.823, and 0.898 for Go2, B2, and G1, respectively. Total variance decreases in eleven of twelve cases, with Go2 backward motion being the only exception at R_{var}=1.225. The D_{W2} values range from 0.252 for Go2 forward motion to 1.404 for Go2 rotational motion, indicating motion-dependent changes under residual control. Together with the decoded representations in Fig. 4 and the tracking MAE in Table II, these results suggest that *NALOCO* modifies contact-related behavior while retaining a broadly consistent motion-dependent representation and the reference policy's velocity-tracking performance.

D. Real-World Acoustic Noise Attenuation

1) Velocity tracking error: On hardware, the body velocity is measured from an Aeva Aeries II 4D LiDAR by averaging point-wise instantaneous velocities inside a body-centered spatial window, and is compared with the commanded v_x.

TABLE III
REAL-WORLD TRACKING ERROR: MEAN±STD TRACKING ERROR [M/S].

Robot		v_x [m/s]				
		0.2	0.4	0.6	0.8	1.0
Go2	w/o res.	0.09±0.02	0.04±0.03	0.06±0.05	0.04±0.02	0.09±0.06
	w/ res.	0.06±0.02	0.04±0.02	0.03±0.02	0.03±0.03	0.06±0.05
B2	w/o res.	0.06±0.04	0.04±0.02	0.05±0.02	0.05±0.03	0.11±0.01
	w/ res.	0.04±0.03	0.01±0.02	0.04±0.02	0.01±0.02	0.08±0.01
G1	w/o res.	0.09±0.04	0.07±0.04	0.04±0.04	0.07±0.04	0.12±0.07
	w/ res.	0.07±0.03	0.03±0.03	0.04±0.04	0.06±0.03	0.06±0.04

Go2. As shown in Table III, the residual lowers the error at 0.2, 0.6, 0.8, and $1.0\,\mathrm{m/s}$ and matches the reference at $0.4\,\mathrm{m/s}$. **B2.** The tracking error decreases at every commanded speed, by 0.01–0.04 m/s. **G1.** Except for $0.6\,\mathrm{m/s}$, where both means are $0.04\,\mathrm{m/s}$, the error is decreased by the residual at every commanded speed. The reductions are 0.01–0.06 m/s. Consistent with Table II, the residual maintains or improves tracking performance on hardware as in MuJoCo.

2) Real-world noise level: Fig. 5 shows the real-world acoustic noise levels while walking on five surfaces—asphalt, grass, gravel, marble, and rubber—under the built-in Model Predictive Controller (MPC), residual-off, and residual-on settings at a commanded forward speed of $1.0\,\mathrm{m/s}$, except for G1's MPC controller which is set to its maximum speed ($\approx 0.33\,\mathrm{m/s}$). Residual-on lowers both peak and mean noise levels relative to residual-off across all robots and terrains. **Go2.** The largest peak reduction relative to residual-off is 5.3 dB on asphalt, from 63.3 to 58.0 dB. This peak is also 5.3 dB lower than that of MPC. The largest mean noise reduction is 5.0 dB on gravel. **B2.** B2 exhibits the largest reductions among the three robots. Relative to MPC, peak noise decreases by 16.2 dB on asphalt, 13.6 dB on grass, and 9.3 dB on marble, with corresponding mean reductions of 12.5 dB on asphalt and 11.2 dB on grass. Relative to residual-

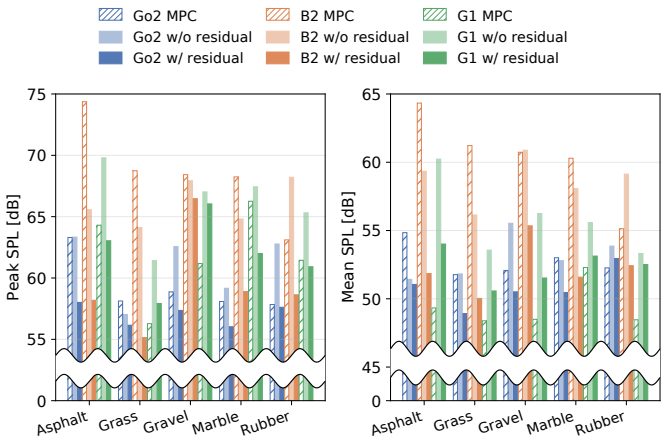


Fig. 5. Real-world SPL on five terrains. Left: peak; right: mean. Hatched bars are MPC, light bars are residual off, and dark bars are residual on.

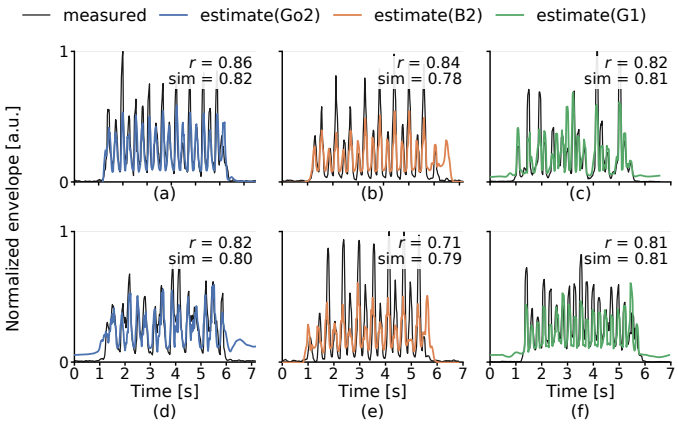


Fig. 6. Microphone envelope and the noise proxy estimated by NPNet during forward and backward walking on flat terrain with the residual policy enabled. (a) Go2, (b) B2, and (c) G1 walk forward, and (d) Go2, (e) B2, and (f) G1 walk backward.



Fig. 7. Residual-enabled locomotion of Go2 on an iron plate, B2 on a grassy slope, and G1 descending a staircase.

off, peak noise decreases by 9.6 dB on rubber, 9.0 dB on grass, and 7.4 dB on asphalt. Mean noise also decreases by 7.5 dB on asphalt. **G1.** Relative to residual-off, peak noise decreases by 6.8 dB on asphalt and 5.4 dB on marble, while mean noise decreases by 6.2 dB on asphalt. On marble, peak noise is also 4.3 dB lower than that of MPC. Overall, these reductions in peak and mean noise levels are consistent with the lower touchdown impacts observed in MuJoCo.

3) Noise Proxy and Real Acoustic Noise: We evaluate the similarity between the noise proxy estimated by NPNet on the robot and the acoustic signal measured by a microphone. Proprioceptive observations and audio are recorded simultaneously on Go2, B2, and G1 while walking on flat terrain with the residual policy enabled. The microphone signal is converted into a log1p-RMS envelope, and the five proxy terms in Eq. (8) are standardized per robot and averaged. To compensate for the fixed latency between the two streams, we align them using a single time lag per recording, selected by maximizing cross-correlation, and then z-normalize the aligned signals. We report their pointwise Pearson correlation r and a dynamic time warping (DTW) similarity score. DTW uses the absolute pointwise difference, and we define nDTW as the accumulated DTW cost divided by the warping-path length. The similarity score is then computed as $\mathrm{sim} = \exp(-\mathrm{nDTW})$, where values closer to one indicate more similar temporal patterns. Fig. 6 shows the measured microphone envelope and the noise proxy estimated by NPNet during forward and backward walking. Across all robots and directions, the proxy tracks the measured peak sequence, with r ranging from 0.71 to 0.86 and sim from 0.78 to 0.82. These results indicate that the estimated noise proxy effectively captures the temporal variations in actual impact-induced noise across different robot morphologies and walking directions. Fig. 7 demonstrates the robustness of the proposed method through real-world locomotion experiments across diverse terrains with the residual activated.

E. Ablation Study

We ablate the noise-proxy components extracted by NPNet, evaluating each configuration while keeping all other settings unchanged. NP-0 excludes the noise proxy, NP-3 uses the three-term proxy defined by Eqs. (3), (4), and (7), and NP-5 uses the full five-term proxy defined by Eq. (8). To assess the effect of reward-based policy optimization without the proposed noise-proxy formulation, π'_{ref} fine-tunes the reference policy using the rewards defined in Eq. (12), whereas π''_{ref} is trained from scratch using the same rewards. As shown in Table IV, NP-5 achieves the lowest impact across all robots and speeds. Increasing the number of proxy terms generally reduces impact, with the clearest improvement observed on B2. Relative to π_{ref}, the maximum condition-wise impact reduction is 15.8% for π'_{ref} and 17.5% for π''_{ref}, compared with 65.68% for NP-5. Table V shows that NP-5 maintains a mean tracking error no greater than that of π_{ref} in all evaluated conditions, although tracking error does not improve monotonically as proxy terms are added. In contrast, training π''_{ref} from scratch substantially increases tracking error at 1.0 m/s on Go2, B2, and G1, despite its modest impact reductions. Fine-tuning π'_{ref} avoids this high-speed degradation and achieves a lower tracking error than NP-5 on G1 at 1.0 m/s, but it does not match NP-5's impact reduction.

V. Conclusion

We presented *NALOCO*, a residual reinforcement-learning framework that attenuates footstep noise through a proprioceptive NPNet latent while keeping the reference locomotion

TABLE IV
NPNET ABLATION: MEAN±STD TOUCHDOWN IMPACT [N].

Robot		Impact [N]				
		0.2 m/s	0.4 m/s	0.6 m/s	0.8 m/s	1.0 m/s
Go2	π_{ref}	34.2±9	53.0±21	87.1±38	96.3±33	108.6±44
	π'_{ref}	33.0±9	51.9±23	81.5±38	89.0±34	108.5±42
	π''_{ref}	33.4±8	50.3±18	80.9±26	92.3±31	106.4±38
	NP-0	32.5±11	46.5±23	85.3±27	90.7±42	98.3±58
	NP-3	30.4±9	46.2±26	75.3±31	87.8±40	96.1±42
	NP-5	26.1±8	41.2±21	67.9±29	75.2±30	93.2±34
B2	π_{ref}	367.1±64	547.9±68	457.4±97	572.0±13	598.8±31
	π'_{ref}	333.6±18	538.7±47	428.5±170	538.7±188	590.2±252
	π''_{ref}	354.2±33	466.7±54	427.7±95	536.8±130	591.0±167
	NP-0	217.7±78	290.6±35	400.7±113	531.2±224	594.8±231
	NP-3	150.2±90	224.2±169	333.7±149	462.2±164	577.4±194
	NP-5	126.0±44	197.6±189	288.4±152	404.7±137	541.0±155
G1	π_{ref}	143.9±24	138.2±32	165.7±19	186.1±42	182.0±46
	π'_{ref}	121.2±34	129.9±46	150.4±51	172.4±43	172.4±54
	π''_{ref}	118.7±63	135.6±66	151.6±75	176.1±77	173.9±83
	NP-0	119.7±48	133.3±33	141.6±11	128.7±57	160.4±90
	NP-3	117.8±34	123.8±59	143.1±52	123.4±47	164.7±68
	NP-5	113.1±52	119.9±34	136.8±62	118.5±74	140.3±82

TABLE V
NPNET ABLATION: MEAN±STD TRACKING ERROR [M/S].

Robot		v_x [m/s]				
		0.2	0.4	0.6	0.8	1.0
Go2	π_{ref}	0.04±0.02	0.03±0.03	0.03±0.02	0.03±0.03	0.04±0.02
	π'_{ref}	0.05±0.03	0.03±0.02	0.03±0.02	0.02±0.03	0.03±0.03
	π''_{ref}	0.05±0.04	0.06±0.02	0.13±0.04	0.14±0.02	0.16±0.04
	NP-0	0.03±0.02	0.02±0.01	0.02±0.02	0.02±0.03	0.03±0.02
	NP-3	0.03±0.03	0.03±0.02	0.03±0.02	0.02±0.03	0.03±0.02
	NP-5	0.03±0.03	0.03±0.03	0.03±0.02	0.02±0.03	0.03±0.03
B2	π_{ref}	0.08±0.06	0.07±0.02	0.06±0.03	0.07±0.04	0.06±0.05
	π'_{ref}	0.04±0.02	0.07±0.03	0.06±0.02	0.03±0.01	0.05±0.02
	π''_{ref}	0.03±0.01	0.04±0.03	0.07±0.03	0.13±0.03	0.18±0.09
	NP-0	0.02±0.02	0.05±0.02	0.05±0.01	0.06±0.04	0.06±0.04
	NP-3	0.02±0.04	0.04±0.04	0.05±0.02	0.04±0.01	0.05±0.02
	NP-5	0.03±0.05	0.03±0.03	0.04±0.01	0.04±0.02	0.04±0.05
G1	π_{ref}	0.10±0.05	0.08±0.04	0.06±0.03	0.11±0.02	0.16±0.02
	π'_{ref}	0.09±0.04	0.08±0.03	0.06±0.02	0.08±0.01	0.07±0.04
	π''_{ref}	0.10±0.06	0.11±0.09	0.07±0.03	0.13±0.03	0.28±0.08
	NP-0	0.07±0.02	0.07±0.03	0.04±0.03	0.08±0.02	0.07±0.02
	NP-3	0.08±0.03	0.08±0.02	0.04±0.03	0.08±0.01	0.08±0.02
	NP-5	0.08±0.03	0.08±0.03	0.05±0.04	0.08±0.02	0.12±0.02

controller frozen. Across quadrupedal and humanoid robots, the residual reduces touchdown impact force and generally yields fewer contacts and longer contact durations while maintaining or improving velocity tracking in simulation and hardware experiments. On five real-world terrains, peak and mean SPL decrease by up to $9.6\,\text{dB}$ and $7.5\,\text{dB}$ relative to residual-off, and by up to $16.2\,\text{dB}$ and $12.5\,\text{dB}$ relative to the built-in MPC. Future work will extend the framework to exteroceptive reference controllers for safe, noise-attenuated locomotion in diverse human-centric environments.

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